Essay 2: Missing Data Strategies

Missing data is one of those problems that looks trivial on the surface, yet it has the power to reshape a model's behaviour in ways that only become obvious later. Géron (2022) makes it clear that there is no neutral choice here—every missing-data strategy carries an assumption, and those assumptions influence what the model ultimately learns. The core approaches fall into deletion, imputation, prediction-based filling, and hybrids such as adding missingness indicators. The key is understanding when each method makes sense and the kind of distortion it introduces.

1. Removing rows with missing values

The simplest approach is to drop any row containing missing data. This can be acceptable when the proportion of missingness is genuinely small and the dataset is large enough that losing a few rows has no meaningful impact. But this only works well when the values are Missing Completely At Random (MCAR). In practice, that assumption rarely holds, and removing such rows can erase meaningful patterns.

2. Removing columns with missing values

Dropping a feature entirely is appropriate when the missingness is extremely high and the feature adds little even when present. However, dropping a feature removes its signal entirely. If we drop the satisfaction column, for example, we lose a valuable target variable (although this depends on what the target is), and we also remove any possibility of the model learning how satisfaction interacts with arrival patterns or service types.

3. Traditional imputation (mean, median, mode)

Imputation is often the most balanced first step. Scikit-learn's SimpleImputer supports mean, median, or mode imputation, which preserves row count but compresses natural variance. Géron notes that this can give models an overly neat version of reality.

4. Imputation with a missingness indicator

Pairing imputation with a binary flag identifying where data was missing preserves both structure and signal. In many datasets, missingness is itself informative—for example, ICU patients may systematically lack satisfaction scores.

5. Prediction-based imputation

More advanced imputers such as KNNImputer or IterativeImputer reconstruct missing values using patterns across the dataset. These approaches can be powerful but carry a larger leakage risk if not wrapped inside a Pipeline.

Example where the choice changes everything:

In a Nigerian hospital dataset where satisfaction is missing mostly for ICU patients, dropping rows erases ICU cases, dropping the column erases an important relationship, mean imputation washes out group differences, and imputation plus a missingness indicator preserves both the value and the fact that it was missing.

References:

Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow. 3rd ed. O’Reilly Media.